

Data Quality and Golden Dataset Report

The Golden Dataset establishes a mathematically sound foundation for operational metrics by systematically eliminating synthetic records, resolving fragmented entities, and enforcing strict attribution rules.

Data Transformation Funnel

Dataset	Raw Records (Est.)	Rejected/Corrected	Golden Dataset Impact
Accounts	Baseline	Dropped 455 orphaned rows	Removed fake denominator volume
Borrowers	30,600	Consolidated 24 duplicates	Prevented entity fanning
Payments	Baseline	Removed 265 unverified rows	Established 23.52% true recovery rate
Vendor Dispositions	Baseline	Excluded August 2026 cohort	Removed late-arriving data bias
Attributions	14,267 matches	Excluded 12,714 organic payments	Prevented 89% ROI overstatement

Source-of-Truth Decisions

- **Revenue:** The cleaned `payments` table is the exclusive source of truth for financial recovery. Synthetic and mathematically uniform `account_status_history` tables were completely invalidated for performance tracking.
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- **Performance:** True operational success is determined solely by the unified touchpoint ledger matched to successful payments, overriding any agent-selected disposition codes.
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Entity Resolution & Deduplication Logic

- **Borrowers:** Consolidated 24 duplicate borrower profiles sharing identical names and emails into a single chronological master ID to prevent population inflation.
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- **Agents:** Mapped fragmented agent IDs into a unified master ledger to prevent fracturing the "recovery per agent-hour" metric.
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- **Payments:** Deduplicated redundant payment webhooks and ingestion retries to calculate actual cash collected.
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Missing-Data & Timestamp Treatment

- **Orphaned Entities:** Permanently dropped 455 accounts missing `borrower_ids` to ensure all debt is tied to a valid, contactable entity.
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- **Contactability Bias:** Flagged 614 missing phone numbers and 895 missing emails in the borrowers table rather than deleting them, preserving the portfolio denominator while identifying unreachable segments.
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- **Timezone Alignment:** Converted every isolated timestamp column (`event_at`, `interaction_time`, `created_at`, etc.) across all 11 operational tables to standardized UTC to enable accurate chronological merging.
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Payment Attribution & Exclusion Rules

- **Unified Ledger Merge:** Concatenated Calls, SMS, WhatsApp, and Field Visits into a single timeline, utilizing an **asof** backward merge to match payments to the immediately preceding interaction.
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- **The 72-Hour Rule:** Established a strict ≤ 72 -hour attribution window. 12,714 payments occurring outside this window (median 17 days post-contact) were excluded from operational ROI and reclassified as organic self-cures.
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- **August Exclusion:** Purged all August 2026 data due to a severe volume drop (roughly 20% of normal volume) representing a confirmed late-arriving data bias.
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Data-Quality Issues & Historical Changes

- **Definitional Bleed:** Corrected a core schema error where late-stage **NPA** (Non-Performing Asset) statuses were misclassified as baseline demographic risk segments.
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- **Vendor Fraud:** Exposed mathematically impossible, uniform distributions in third-party telephony logs (identical volumes of **PAID**, **NO_CONTACT**, and **WRONG_NUMBER**), proving the disposition data was synthetic or mechanically assigned.
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- **Redundant Coding:** Consolidated **PTP** and **PROMISE_TO_PAY** disposition codes to fix a reporting fragmentation error that was splitting the primary success metric in half.
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Core Analytical Assumptions

- Assumed that identical names and emails in the borrowers table represent the same physical human being.
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- Assumed that payments occurring more than 3 days after a documented operational touchpoint were not directly caused by that touchpoint, requiring separation from the ₹10 Cr investment ROI model.